# AquaSense X200 Technical Manual

***Water Quality IoT Monitor | Model AS-X200 | Firmware 2.4.1***

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# 1. Product Overview

The AquaSense X200 is an IoT water-quality monitoring device for industrial and municipal use. It measures pH, turbidity, temperature, dissolved oxygen (DO) and total dissolved solids (TDS), and sends readings to the NovaCloud dashboard over Wi-Fi or 4G LTE. Firmware version covered in this manual: 2.4.1. Model number: AS-X200. Warranty: 24 months from date of purchase.

# 2. Technical Specifications

| Parameter | Specification |
|---|---|
| pH range / accuracy | 0 to 14 pH / +/- 0.1 pH |
| Turbidity range | 0 to 1000 NTU, accuracy +/- 2% |
| Temperature range | 0 to 80 degrees C, accuracy +/- 0.5 degrees C |
| Dissolved oxygen | 0 to 20 mg/L, accuracy +/- 0.3 mg/L |
| TDS | 0 to 5000 ppm |
| Power supply | 12 V DC, 2 A adapter; backup battery 3000 mAh (about 8 hours) |
| Connectivity | Wi-Fi 802.11 b/g/n (2.4 GHz), 4G LTE, Bluetooth 5.0 for setup |
| Ingress protection | IP67 (sensor probe), IP54 (main unit) |
| Operating temperature | -10 to 50 degrees C |
| Sampling interval | Configurable: 10 seconds to 60 minutes (default 5 minutes) |
| Weight | 1.8 kg |

# 3. Box Contents

- AquaSense X200 main unit
- Multi-parameter sensor probe with 3 m cable
- 12 V DC power adapter
- Wall-mount bracket and 4 screws
- pH calibration solutions (pH 4.00, 7.00, 10.00), 50 mL each
- Quick start guide

# 4. Installation

## 4.1 Mounting

- Choose a location within 3 m of the water source and within Wi-Fi range (or with 4G signal).
- Fix the wall-mount bracket using the 4 screws provided, at least 1.2 m above ground.
- Slide the main unit onto the bracket until it clicks.
- Submerge the probe at least 10 cm below the water surface. Keep it away from air bubbles and the tank walls.

### 4.2 First-time Setup

• Connect the 12 V adapter and press the power button for 3 seconds. The LED turns solid blue.

• Install the NovaSense mobile app and pair via Bluetooth. The default pairing PIN is 0000.

• Select your Wi-Fi network and enter the password. The LED blinks green while connecting and turns solid green when online.

• Register the device on the NovaCloud dashboard using the 12-character serial number printed on the back.

## 5. Calibration

Calibrate the pH sensor every 30 days, or sooner if readings drift. Use fresh buffer solutions at room temperature (25 degrees C).

• Rinse the probe with distilled water and blot dry.

• Open NovaSense app, go to Settings, Calibration, pH.

• Immerse the probe in pH 7.00 solution and wait 60 seconds for the reading to stabilise. Tap "Set Point 1".

• Rinse, then immerse in pH 4.00 (or 10.00) solution. Tap "Set Point 2".

• The app shows "Calibration OK" if the slope is between 90% and 110%. Otherwise replace the sensor.

Turbidity and TDS sensors are factory-calibrated and need verification once every 6 months. The DO sensor membrane should be replaced every 12 months.

## 6. LED Status Indicators

| LED | Meaning | Action |
|---|---|---|
| Solid blue | Powered on, waiting for setup | Pair with the app |
| Blinking green | Connecting to network | Wait up to 60 seconds |
| Solid green | Online, working normally | None |
| Blinking amber | Low battery (below 15%) | Connect the power adapter |
| Solid red | Sensor fault | Check probe cable; restart device |
| Blinking red | Firmware update failed | Perform factory reset |

## 7. Troubleshooting

| Problem | Likely Cause | Solution |
|---|---|---|
| Device will not power on | Faulty adapter or loose connector | Try another 12 V / 2 A adapter; press power button for 10 seco |
| No data on dashboard | Wi-Fi password changed or weak signal | Re-run Wi-Fi setup in app; move closer to router or enable 4G |
| pH reading unstable | Dirty probe or expired buffer | Clean probe with soft brush and recalibrate |
| Turbidity always 0 NTU | Probe not submerged or lens blocked | Submerge at least 10 cm and wipe lens |

| Problem | Likely Cause | Solution |
|---|---|---|
| Error code E04 | Temperature sensor open circuit | Reconnect probe cable; replace probe if it persists |
| Error code E11 | Calibration slope out of range | Recalibrate with fresh buffers |
| Error code E27 | Firmware corrupted | Hold reset button 15 seconds for factory reset, then reinstall firm |

## 8. Maintenance Schedule

| Task | Frequency |
|---|---|
| Clean sensor probe | Every 2 weeks |
| pH calibration | Every 30 days |
| Verify turbidity and TDS | Every 6 months |
| Replace DO membrane | Every 12 months |
| Replace backup battery | Every 24 months |

## 9. Firmware Update and Factory Reset

Firmware updates are delivered over the air. In the NovaSense app, go to Settings, Device, Check for Updates. Do not power off during the update, which takes about 4 minutes. To factory reset, hold the reset button (small hole on the right side) for 15 seconds until the LED blinks red twice. A reset erases Wi-Fi credentials and calibration data but not cloud history.

## 10. Safety and Support

- Do not use the device in flammable or explosive atmospheres.
- Disconnect power before cleaning the main unit. Do not immerse the main unit.
- Dispose of the battery as e-waste according to local regulations.

Technical support: support@novatech.example, phone +91 40 5555 0199, Monday to Saturday 9 AM to 7 PM IST. Please keep the serial number ready.